Saturated Superfluid Vacuum Goldstone

The Photon as a Goldstone Mode and the Chiral-Shear Near Field

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Abstract

Paper II of the Saturated Superfluid Vacuum (SSV) series identified a tension in the electromagnetic sector. The static Coulomb and magnetic near-field mechanisms arise naturally from chiral-shear flow gradients around toroidal charged vortices, with coupling strength set by the dimensionless parameter α. But the same preliminary derivation identified the propagating photon with the chiral-shear mode at $c_\perp$, producing a speed mismatch: observed light in vacuum propagates at c, not at $c_\perp$.

This paper isolates the correction required by that audit. The observed photon should be identified with the Goldstone/Bogoliubov phase mode of the saturated order parameter Ψ, propagating at the medium sound speed c. The chiral-shear sector remains physically real, but it is reassigned to the static and near-field electromagnetic structure: Coulomb pressure, magnetic circulation, Aharonov–Bohm phase, and vortex-ring self-energy.

The central task is therefore a two-channel derivation. One must derive the radiative Maxwell sector from the Goldstone phase channel at speed c, while preserving the static Coulomb and magnetic results of Paper II as the near-field limit of chiral-shear topology. We do not claim this derivation is complete here. We state the consistency conditions, identify the matching problem between the two channels, and define the success criterion for a future full calculation.

Update (SSV issue 138, CHIRAL-DISPERSION). The channel question has since been settled by a pre-registered computation: linearising the full LogSE + chiral-shear + CP^1 functional around the uniform vacuum shows the chiral-shear term is *silent* in the linear spectrum (the linear-order current is a pure gradient; the $|\nabla \times \mathbf{j}|^2$ energy is quartic), so there is no propagating chiral-shear mode at any speed — and the internal-direction channel disperses as $\omega = \hbar k^2/2m_0$, gapless, quadratic, charge-decoupled at linear order. The Goldstone assignment proposed here is therefore no longer one candidate among two: it is the only propagating gapless linear channel the medium has. The photon's two polarizations are identified with the internal CP^1 direction transported on the phase channel; deriving that transport (the matching programme of this note, with displacement-current coefficient $1/c^2$) is the remaining obligation.

Contents

1 Introduction

The electromagnetic sector of SSV has split into two logically distinct pieces:

(i) a *static near-field sector*, where charge, Coulomb force, magnetic circulation, and AB phase arise from the topology and chiral-shear flow of toroidal vortices;

(ii) a *radiative sector*, where the observed photon must propagate at the measured vacuum speed c.

Paper II developed the first piece but used the wrong carrier for the second. It treated the chiral-shear mode as the photon and assigned it the speed $c_\perp$. This cannot be the final physical interpretation because the observed photon propagates at c.

The correction is suggested by analogue-gravity and logarithmic-superfluid vacuum theory: the physical photon is the Goldstone phase mode of Ψ [?, ?]. The phase mode propagates at the medium sound speed,

$$c_s(\rho_0) = c, \tag{1}$$

and is therefore compatible with observed light propagation and with gravitational-wave constraints such as GW170817.

The question is not simply whether one can rename the photon. The hard question is whether the successful static electromagnetic mechanisms of Paper II survive the replacement:

$$\text{chiral-shear photon at } c_\perp \quad \longrightarrow \quad \text{Goldstone photon at } c.$$

The present paper formulates that problem.

Claim of this paper. The observed photon should be the Goldstone/Bogoliubov phase mode of the SSV order parameter, propagating at c. The chiral-shear field should be retained as the static near-field sector that fixes Coulomb strength and magnetic topology. The open calculation is the matching between these two sectors.

2 The Problem in Paper II

Paper II recovered a set of electromagnetic structures from chiral vortex topology:

$$F_C(r) = \frac{\alpha\hbar c}{r^2}, \tag{2}$$

$$\mathbf{B} \sim \nabla \times \mathbf{v}_\perp, \tag{3}$$

$$\gamma_{\mathrm{AB}} = \frac{q}{\hbar}\oint \mathbf{A}\cdot d\mathbf{l}. \tag{4}$$

Here $\mathbf{v}_\perp$ is the chiral-shear flow associated with the toroidal vortex, and α is the empirical dimensionless strength of that sector.

The same section then tentatively identified the propagating photon with a transverse chiral-shear phonon satisfying

$$\partial_t^2 \delta\theta = c_\perp^2 \nabla^2 \delta\theta, \qquad c_\perp \neq c. \tag{5}$$

This produces the speed problem. If the physical photon propagated at $c_\perp$, vacuum light speed would be $c_\perp$, not c. The earlier patch that $\alpha \to 1$ in free vacuum is not available inside SSV because vacuum *is* the saturated medium.

The diagnosis is therefore:

$$\text{static chiral-shear mechanism: keep,} \qquad \text{radiative photon identification: replace.} \tag{6}$$

The SSV issue 138 computation subsequently showed that equation (**??**) was not merely the wrong photon — it does not arise from the functional at all: around the uniform vacuum the chiral-shear term contributes nothing at linear order, and the internal-direction channel it was thought to describe is first-order in time with $\omega = \hbar k^2/2m_0$. The "slow second light" has no linear mode behind it; the diagnosis above is thereby upgraded from an audit conclusion to a computed result.

3 Goldstone Channel of the Saturated Medium

Write the order parameter as

$$\Psi(\mathbf{x},t) = \left(\sqrt{\rho_0} + \frac{\delta\rho}{2\sqrt{\rho_0}}\right) e^{i\theta_0 + i\delta\theta}. \tag{7}$$

Linearisation gives two familiar modes:

- the amplitude mode $\delta\rho$, gapped by the stiffness of the saturated medium;
- the phase mode $\delta\theta$, the Goldstone/Bogoliubov mode.

At long wavelengths, the phase mode obeys

$$\partial_t^2 \delta\theta = c^2 \nabla^2 \delta\theta, \tag{8}$$

with dispersion

$$\omega = ck. \tag{9}$$

This is the natural carrier for physical radiation in SSV. It is massless, universal, and propagates at the observed vacuum light speed. It is also the same channel used in the acoustic-metric reading of gravity, where local density variations modify the local propagation rate.

> **Channel assignment.** The Goldstone/Bogoliubov channel carries propagating radiation at c. The chiral-shear channel fixes the handed near-field structure of charged defects and the coupling strength α.

4 Two-Channel Electromagnetism

The corrected EM sector should be written as a two-channel theory:

$$\text{near field:} \quad \mathbf{v}_\perp,\ \Phi_\perp,\ \mathbf{A}_\perp, \qquad \text{chiral-shear/topological sector,} \tag{10}$$

$$\text{radiation:} \quad \delta\theta_{\mathrm{G}}, \qquad \text{Goldstone phase channel.} \tag{11}$$

The near field knows about charge chirality and the dimensionless coupling α. The radiative field knows about propagation at c.

This is not unusual from the point of view of ordinary electrodynamics. In QED, the Coulomb field is not a freely propagating transverse photon; it is the constrained or longitudinal part of the gauge field. Radiation is carried by transverse on-shell photon modes. SSV should reproduce this division mechanically:

$$\text{Coulomb interaction} \leftrightarrow \text{static chiral-shear pressure constraint,} \tag{12}$$

$$\text{on-shell light} \leftrightarrow \text{Goldstone phase wave at } c. \tag{13}$$

The task is to derive the effective Maxwell system in which the source sector is chiral-shear/topological, while the homogeneous radiative sector propagates at c:

$$\nabla \cdot \mathbf{E} = \rho_{\rm ch}/\epsilon_0, \tag{14}$$
$$\nabla \cdot \mathbf{B} = 0, \tag{15}$$
$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \tag{16}$$
$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2}\partial_t \mathbf{E}. \tag{17}$$

Relative to the provisional Paper II derivation, the displacement-current coefficient must be $1/c^2$ rather than $1/c_\perp^2$.

5 What Must Survive

The Goldstone correction is acceptable only if it preserves the successful static and topological content of Paper II. The following structures must survive.

Coulomb strength. The static force must remain

$$F_C(r) = \frac{\alpha \hbar c}{r^2}. \tag{18}$$

In the corrected reading, α is not a photon-speed ratio. It is the strength of the chiral-shear near-field coupling.

Magnetic topology. The magnetic field must remain the curl of a vector flow,

$$\mathbf{B} \propto \nabla \times \mathbf{v}_\perp, \tag{19}$$

so that $\nabla \cdot \mathbf{B} = 0$ remains a vector identity and magnetic monopoles remain topologically forbidden.

Aharonov–Bohm phase. The phase shift must remain a Berry phase of the charged ring in a background circulation:

$$\gamma_{\rm AB} = \frac{q}{\hbar} \oint \mathbf{A} \cdot d\mathbf{l}. \tag{20}$$

This is naturally near-field/topological rather than radiative.

Anomalous magnetic moment. The one-loop self-energy picture must be reinterpreted carefully. The virtual exchange responsible for $g - 2$ may continue to involve the chiral-shear near field, but the on-shell photon propagator must belong to the Goldstone channel. The calculation therefore has to distinguish virtual constrained exchange from physical radiative quanta.

> **Open matching problem.** The central technical challenge is to show that the chiral-shear near field and the Goldstone radiation field are not two unrelated electromagnetic fields, but two limits of one effective EM response of the saturated medium.

6 The Matching Calculation

A complete derivation should start from the full SSV functional,

$$S[\Psi] = \int dt\, d^3x \, [\mathcal{L}_{\mathrm{LogSE}}(\Psi) + \mathcal{L}_{\perp}(\Psi, \nabla\Psi) + \mathcal{L}_{\mathrm{defect}}] \, , \tag{21}$$

expand around a charged toroidal background Ψ_{e}, and separate fluctuations into constrained near-field and propagating Goldstone parts:

$$\Psi = \Psi_{\mathrm{e}} + \delta\Psi_{\mathrm{near}} + \delta\Psi_{\mathrm{rad}}. \tag{22}$$

The desired effective action has the schematic form

$$S_{\mathrm{eff}} = \int dt\, d^3x \left[\frac{\epsilon_0}{2} E^2 - \frac{1}{2\mu_0} B^2 + J^{\mu} A_{\mu} \right] , \tag{23}$$

with two requirements:

$$\text{static limit:} \quad \nabla^2 \Phi = -\rho_{\mathrm{ch}}/\epsilon_0, \qquad F_C = \alpha\hbar c/r^2, \tag{24}$$

$$\text{radiative limit:} \quad \partial_t^2 A_i - c^2 \nabla^2 A_i = 0, \qquad \omega = ck. \tag{25}$$

The calculation is therefore a matched asymptotic problem. Near charged defects, chiral-shear topology fixes the source and static response. Far from sources, the propagating disturbance must project onto the Goldstone phase mode.

7 Avoiding New Circularity

The Goldstone photon correction must not simply insert Maxwell theory by hand. The following shortcuts would not count as derivations:

(i) postulating a separate electromagnetic gauge field in addition to Ψ;

(ii) replacing $c_{\perp}$ by c in Paper II's equations without deriving the channel projection;

(iii) keeping Coulomb from chiral shear and radiation from Goldstone without deriving their matching at sources;

(iv) using observed light speed to tune a coefficient in the chiral-shear sector.

A valid derivation should show that the same saturated order parameter supports both the constrained static response and the propagating radiation response, with no additional gauge field inserted by hand.

Success criterion. The Goldstone photon programme succeeds if the full SSV linearisation around charged toroidal defects yields Maxwell electrodynamics with static Coulomb strength $\alpha\hbar c$, magnetic topology from chiral-shear circulation, and on-shell radiation satisfying $\omega = ck$.

8 Consequences

If the matching calculation succeeds, several tensions in Paper II are resolved at once.

Light and gravity share the correct speed. Both electromagnetic radiation and gravitational disturbances can be read as Goldstone/Bogoliubov-channel excitations at c, consistent with GW170817.

α is not a speed ratio. The fine-structure constant becomes a chiral-shear/topological coupling and defect-geometry parameter, not a ratio involving the observed photon speed. This matches the separate alpha-paper proposal $\alpha = \xi/R^*$.

The old EM derivation becomes a near-field derivation. The Paper II chiral-shear equations are not discarded; they are reclassified. They describe constrained near-field structure and virtual exchange rather than the on-shell photon.

Magnetic monopoles remain forbidden. Because the magnetic field still descends from a curl of the chiral-shear flow, $\nabla \cdot \mathbf{B} = 0$ remains topological rather than postulated.

9 Discussion

The Goldstone photon problem is a cleanup calculation, but it is not a minor one. It decides whether the SSV electromagnetic sector can be made internally coherent after the propagation-speed audit. The desired result is conceptually simple: sources live in chiral topology; radiation lives in the Goldstone phase channel. The hard part is proving that these are two faces of one effective electromagnetic field.

This also clarifies the status of Paper II. Its static Coulomb and magnetic results remain valuable because they identify how charge and magnetic topology arise mechanically. Its original radiative photon identification should be treated as provisional scaffolding. The present paper states the corrected target.

10 Conclusion

The observed photon in SSV should be the Goldstone/Bogoliubov phase mode of the saturated order parameter, propagating at c. The chiral-shear sector should be retained as the near-field and topological electromagnetic sector, where it fixes Coulomb strength, magnetic circulation, and the role of α.

The open derivation is the matching calculation:

$$\boxed{\text{chiral-shear near field} \quad \longleftrightarrow \quad \text{Goldstone radiation at } c}$$

If this can be derived from the full SSV functional without adding a separate gauge field, the electromagnetic sector becomes substantially more coherent. If it cannot, the SSV force-sector programme must be revised at one of its central joints.

References

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